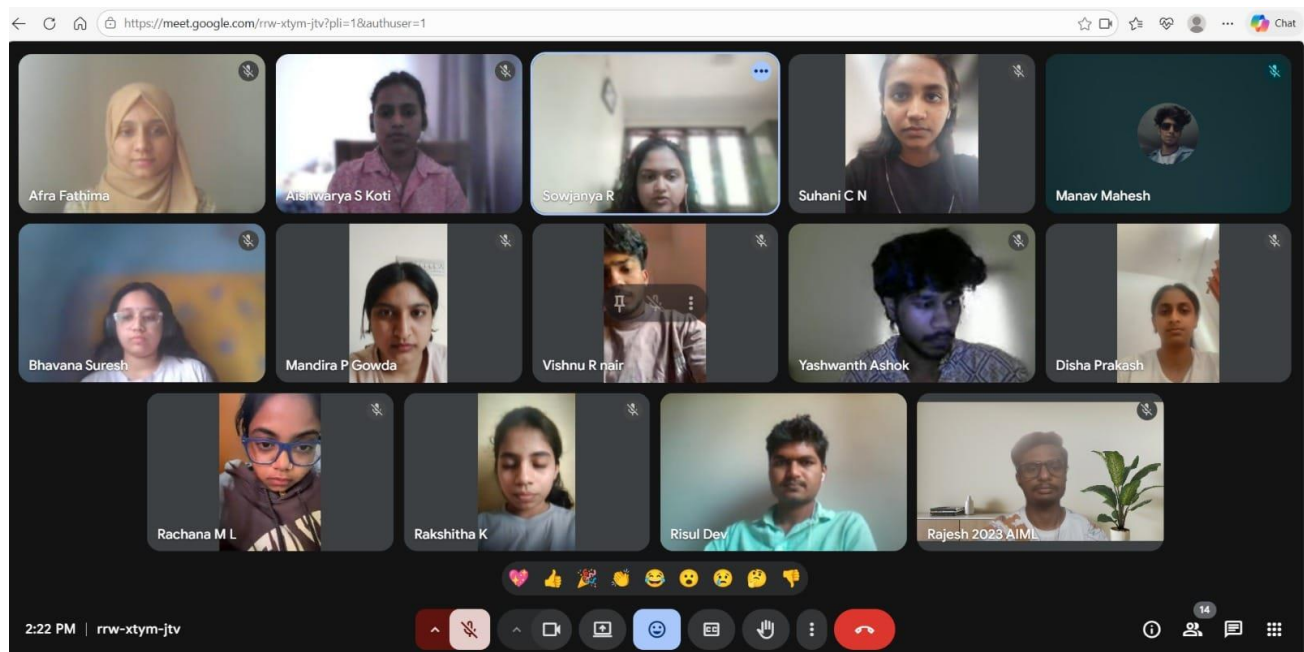


Mini Project: Cohort 21: Forest Fire Ignition Risk

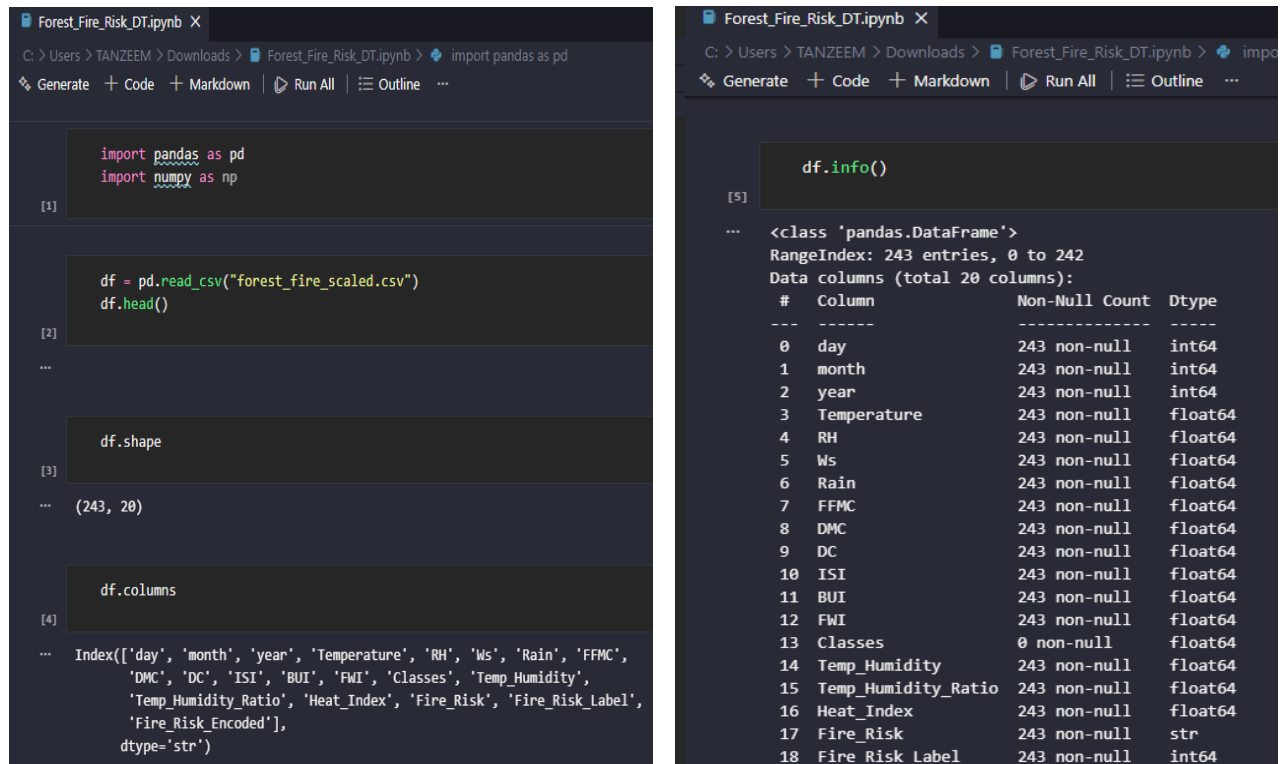
Date:09/03/2026

On Day 3 of the project, the team moved forward to the **model building phase**, which is a crucial step in any machine learning project. In the previous stage, the data collection team had already provided the cleaned dataset, allowing the model building team to begin their work without any delays. The dataset selected for the project is the **Algerian Forest Fire Dataset**, which contains different environmental and meteorological attributes that help in identifying the level of forest fire risk. Since the dataset was already organized and prepared earlier, the team could directly start developing predictive models using the available data.



At the beginning of the discussion, the team talked about which machine learning algorithms would be suitable for solving the problem. Since the project focuses on predicting the risk level of forest fires, it can be treated as a classification problem. After discussing different possible algorithms, the team decided to experiment with two models: **Decision Tree** and **Random Forest**. Both of these models are widely used for classification tasks and are known to perform well on structured datasets.

On this day, the team mainly focused on starting the implementation of the Decision Tree model. A decision tree works by dividing the dataset into smaller groups based on certain conditions related to the features in the data. These conditions form a tree-like structure where each node represents a decision and each branch represents a possible outcome. This process continues until the model reaches a final prediction.



```
[1] import pandas as pd
import numpy as np

[2] df = pd.read_csv("forest_fire_scaled.csv")
df.head()

...

[3] df.shape

...
(243, 20)

[4] df.columns

...
Index(['day', 'month', 'year', 'Temperature', 'RH', 'Ws', 'Rain', 'FFMC',
      'DMC', 'DC', 'ISI', 'BUI', 'FWI', 'Classes', 'Temp_Humidity',
      'Temp_Humidity_Ratio', 'Heat_Index', 'Fire_Risk', 'Fire_Risk_Label',
      'Fire_Risk_Encoded'],
      dtype='str')
```

```
[5] df.info()

...
<class 'pandas.DataFrame'>
RangeIndex: 243 entries, 0 to 242
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   day                    243 non-null   int64
1   month                  243 non-null   int64
2   year                   243 non-null   int64
3   Temperature            243 non-null   float64
4   RH                     243 non-null   float64
5   Ws                     243 non-null   float64
6   Rain                   243 non-null   float64
7   FFMC                   243 non-null   float64
8   DMC                    243 non-null   float64
9   DC                     243 non-null   float64
10  ISI                     243 non-null   float64
11  BUI                     243 non-null   float64
12  FWI                     243 non-null   float64
13  Classes                 0 non-null     float64
14  Temp_Humidity           243 non-null   float64
15  Temp_Humidity_Ratio     243 non-null   float64
16  Heat_Index              243 non-null   float64
17  Fire_Risk               243 non-null   str
18  Fire_Risk_Label         243 non-null   int64
```

The team chose the decision tree model because it is simple, easy to understand, and provides clear insights into how predictions are made. It also automatically selects important features while splitting the data, which helps in identifying the most relevant factors affecting forest fire risk.

CODE SNIPPETS:

```
mm_scaler = MinMaxScaler()
df_minmax = df.copy()
df_minmax[cols_to_scale] = mm_scaler.fit_transform(df[cols_to_scale])

print("MinMax Scaled - first 5 rows:")
print(df_minmax[cols_to_scale].head())
print()
print("Min values (should be 0):")
print(df_minmax[cols_to_scale].min())
print("Max values (should be 1):")
print(df_minmax[cols_to_scale].max())
```

```
df_scaled.to_csv("forest_fire_scaled.csv", index=False)
print("Saved: forest_fire_scaled.csv")
print("Shape:", df_scaled.shape)
print("Columns:", df_scaled.columns.tolist())
```

[29]

```
... Saved: forest_fire_scaled.csv
Shape: (243, 20)
Columns: ['day', 'month', 'year', 'Temperature', 'RH', 'Ws', 'Rain', 'FFMC', 'DMC', 'DC',
```

MODEL TRAINING

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
```

[30]

```
print(df.columns)
```

[31]

Act

```
X = df.drop(columns=["Fire_Risk", "Fire_Risk_Label", "Fire_Risk_Encoded", "Classes", "FWI"])
y = df["Fire_Risk_Label"]
```

Python

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
```

Python

```
X_train
```

Python

```
y_train
```

Python

```
accuracy = accuracy_score(y_test, y_pred)
print()
print("Accuracy:", accuracy)
```

```
print("Confusion Matrix:")
print()
print(confusion_matrix(y_test, y_pred))
```

```
print("Classification Report:")
print()
print(classification_report(y_test, y_pred))
```

Along with this, the team also discussed comparing it with the Random Forest model. The plan is to build both models and allow the evaluation team to test their performance and accuracy. Based on the evaluation results, the model that performs better will be selected as the final model for the project